

VÄINÖ RIKKONEN

Infrastructure & Network Architect

Helsinki, Finland · vaino.rikkonen@gmail.com · rikkova.fi · linkedin.com/in/rikkova · github.com/rikkova

PROFESSIONAL SUMMARY

Infrastructure and network architect with 8+ years across hyperscale operations (Google), enterprise datacenter transformation (CGI), and cyber defense (Accenture). Designs and migrates network fabrics, datacenters, and the automation that operates them: VXLAN-EVPN fabric migration with zero service interruption, datacenter consolidation of 20+ racks, SASE/Zero Trust remote access for ~4,000 users, and Ansible/Python automation that cut manual operations by 50% across a 100+ device fleet. Treats infrastructure decisions as business decisions: EOL decommissioning and contract consolidation delivering €50–100k/yr in OpEx reduction. Currently building a self-hosted LLM agent that validates network designs against the team's SOP library.

CORE SKILLS

Network Architecture: BGP, OSPF, VXLAN, EVPN, MPLS, DWDM, BFD, IPv6, routing & switching, Cisco, FortiGate, firewall management, load balancing

Automation & IaC: Ansible, Python, Terraform, Git, configuration management, infrastructure as code, zero-drift operations

Data Center & Platform: Data center migration and consolidation, Nutanix HCI, virtualization, capacity planning, rack/power/cooling design, high availability, disaster recovery

Cloud & Security: SASE, Zero Trust (Zscaler), AWS Direct Connect, Azure ExpressRoute, hybrid cloud connectivity, network segmentation

AI & LLM: Self-hosted LLM operations (since 2023), retrieval-augmented generation (RAG), LLM-assisted network engineering, human-in-the-loop design, hyperscale AI compute infrastructure

Observability: Telegraf, InfluxDB, Grafana (TIG stack), BMP (RFC 7854), NetFlow, network monitoring, synthetic checks

Leadership: Technical leadership, mentoring, architecture decision records, SOP authoring, cross-team coordination, technical interviewing

PROFESSIONAL EXPERIENCE

Infrastructure Architect — CGI

February 2026 – Present · Helsinki, Finland

- Designed and engineered migration of enterprise remote access from a legacy VPN platform to a SASE-based Zero Trust solution (Zscaler) for ~4,000 users; full-tunnel VPN eliminated, admin network segmentation preserved.
- Leading migration of a legacy Cisco UCS estate to a Nutanix HCI platform: architecture, migration planning, network design, and service transition for 1,000+ VMs.
- Architecting a self-hosted LLM agent with RAG over the team's SOP library: validates network designs, answers engineering questions from internal documentation, and reads live network state through the Ansible framework — read-only by design; production rollout planned for 2026.
- Re-engineered shared Internet edge with BFD-based fast failure detection, cutting routing convergence from minutes to seconds for all shared-service customers.

Network Designer — CGI

April 2024 – February 2026 · Helsinki, Finland

- Owned end-to-end strategy for decommissioning a production datacenter and consolidating 20+ racks of critical, multi-vendor infrastructure into existing facilities; zero service interruption, every move within its agreed maintenance window.
- Led migration of a production fabric from spanning-tree to VXLAN-EVPN overlay (300+ VLANs) via phased parallel-run cutover; MPLS integration sustained, failover cut from ~30–50s STP reconvergence to sub-second ECMP reroute.
- Architected and deployed Ansible/Python automation for lifecycle management and configuration changes: 50% fewer manual touchpoints across 100+ multi-vendor devices, multi-device changes cut from hours to minutes.
- Built full-stack telemetry on the TIG stack (Telegraf, InfluxDB, Grafana) with BGP monitoring via BMP and NetFlow at the internet edge.
- Decommissioned 50+ underutilized and EOL devices and eliminated their license and support contracts, cutting annual infrastructure spend by €50–100k (roughly 10–15%).

- Designed resilient architectures using VXLAN, MPLS core connectivity, and DWDM inter-datacenter transport; co-designed rack layouts, power, and cooling with Datacenter Operations.
- Authored the team's SOP library for network deployment, VLAN/VRF provisioning, and multi-vendor upgrades; mentored junior engineers as technical lead for migrations and upgrades.

Data Center Technician — Google

January 2023 – April 2024 · Hamina, Finland

- Operated hyperscale server and network infrastructure — including the compute behind Google's AI workloads; team's subject matter expert for network maintenance under global SLA/SLO requirements, escalating issues with potential to impact global clusters.
- Developed diagnosis and repair documentation adopted across Google's global fleet, streamlining workflows and reducing downtime beyond the local site.
- Deployed new server and network infrastructure ahead of schedule with rigorous QA; complex repairs across servers, optics, fiber, and power infrastructure; 100% compliance in secure data destruction.
- Served as technical interviewer; cross-trained the local team after completing Google's advanced international technical training program.

Data Networking Engineer — CGI

June 2021 – December 2022 · Helsinki, Finland

- Primary technical lead for Cisco datacenter operations and multi-vendor integrations for enterprise clients; escalation point for complex troubleshooting and RMA workflows.
- Directed datacenter projects covering deployment, decommissioning, and physical migration of network assets with zero stakeholder impact.
- Introduced Ansible and Python scripting to routine operations — the foundation of automation frameworks later deployed at fleet scale.
- Deployed and maintained switches, routers, load balancers, and wireless controllers; resolved complex L2/L3 incidents against strict uptime requirements.

Cyber Defense Analyst — Accenture

August 2020 – June 2021 · Helsinki, Finland

- Investigated and remediated high-priority SIEM alerts and security breaches for global enterprise clients; primary technical contact during on-call rotations for critical incidents.
- Built custom detection logic from post-incident analysis; conducted systematic threat hunting for latent vulnerabilities and APTs.

Network Specialist — CGI

February 2019 – August 2020 · Helsinki, Finland

- Provisioned and maintained multi-vendor enterprise networks (Cisco, Fortinet): switches, routers, load balancers, firewalls; established documentation standards for topologies and change management.

IT Technician — CGI

December 2017 – February 2019 · Helsinki, Finland

- Onsite infrastructure support for a 10,000+ employee environment with 99%+ user uptime; installed and configured network edge equipment and led campus hardware rollouts.

SELECTED PROJECTS

- Datacenter Decommissioning & Consolidation (Technical Lead) — rikkova.fi/work/datacenter-migration.html
- VXLAN-EVPN Fabric Migration (Lead Architect) — rikkova.fi/work/vxlan-evpn-migration.html
- SASE / Zero Trust Remote Access, ~4,000 users (Lead Architect) — rikkova.fi/work/sase-zero-trust-migration.html
- Network Automation Framework, 100+ devices (Lead Engineer) — rikkova.fi/work/network-automation-framework.html
- Intelligent Network Agent — self-hosted LLM with RAG over SOP library (Architect · in development, production rollout planned 2026)

EDUCATION & CERTIFICATIONS

Cisco CCNP Service Provider (350-501 SPCOR) — in progress

Google Advanced International Technical Training Program — completed

Vocational Qualification in Business Information Technology

Omnia, Espoo, Finland